

Farhana Najnin Mahi

Data Analyst | AI/ML Researcher | AI Automation Engineer



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CAREER OBJECTIVE

Computer Science and Engineering graduate from Ahsanullah University of Science and Technology with hands-on experience building LLM-orchestrated automation agents (Claude Skills) for live production environments, alongside machine learning, NLP, and full-stack development. Skilled in workflow automation, REST APIs, RAG pipelines, data-pipeline validation, and Power BI dashboards. Interested in building practical, production-grade AI automation systems that create measurable operational impact.

EDUCATION

Ahsanullah University of Science and Technology B.Sc. in Computer Science and Engineering Tejgaon, Dhaka	Nov 2021 - Dec 2025
Dhaka City College Higher Secondary Certificate (HSC) GPA: 5.00 Dhanmondi, Dhaka	2018 - 2020
Dagonbhuiyan Girls High School Secondary School Certificate (SSC) GPA: 5.00 Dagonbhuiyan, Feni	2016 - 2018

EXPERIENCE

Executive, Data Analyst — Foodi Express Limited US-Bangla Group Concern <i>Shops Department · Onsite · Data Analysis · Dashboard Management · Data Handling</i> - Support shop/vendor onboarding by verifying shop details and menu/product listings for accuracy, and identifying duplicate or inconsistent shop entries. - Monitor shop-level performance (order volume, ratings, cancellation rate) to identify best- and worst-performing shops and flag inactive or underperforming shops for follow-up. - Audit product pricing for consistency and accuracy, monitor stock and availability status, and compare pricing against competitors where applicable. - Prepare daily, weekly, and monthly Excel reports, trackers, and pivot-based summaries for KPI monitoring, sales and stock visibility, and management decision-making. - Identify data inconsistencies and frontend-facing errors in pricing, availability, and shop information, and coordinate with shop owners/vendors to correct them. - Track shop-wise revenue and commission figures to support financial visibility and reporting. - Analyze shop-wise performance of promotional and discount campaigns to support marketing and business decisions.	Aug 2026 - Present
Research Intern ELITE Research Lab <i>Remote · AI/ML Research · LLM/NLP · Paper Analysis · Technical Documentation</i> - Assist with AI/ML and LLM/NLP research activities including literature review, paper reading, and research note preparation. - Support experiment planning, result interpretation, and structured documentation for ongoing research tasks. - Contribute to technical reporting by organizing findings, summarizing papers, and maintaining reproducible research records.	Jul 2026 - Present
AI Automation Intern Confidence Infrastructure PLC <i>Scalability Team (AI Automation) · Onsite · Supply Chain Cashflow & SCM Dashboards · LLM-Orchestrated Agents (Claude Skills) · Workflow Automation</i> - Built and piloted two independent Claude-Skills-based automation agents on CIPLC's internal dashboards — a Cashflow Agent for the Supply Chain Cashflow Dashboard and an SCM Agent for the Supply Chain Management (SCM) Dashboard — using Python (pandas, openpyxl, reportlab, python-docx). Cashflow Agent: built cashflow-mismatch forecasting (4/8/12-week horizons) and project-speed scoring across a live portfolio of 50+ EPC/infrastructure tender projects; diagnosed portfolio-wide funding gaps (BDT 12.37Cr at 4 weeks to BDT 65.82Cr at 12 weeks) on 36 of 50 live projects, and validated the pipeline with 26,000+ automated verification checks (0 failures) across 73 source Excel files. SCM Agent: automated PR cut-off tracking (160 uncovered PRs, BDT 10.7M), PR-to-PO conversion and in-transit/overdue shipment monitoring, inventory reorder alerts (278 flagged items, BDT 144.7M reorder value), delivery-status tracking, and monthly	Jun 2026 - Aug 2026

fund/shortage forecasting (BDT 376.4M across Jun-Sep, 1,053 shortage line items), with automated weekly/monthly management reporting.

- **SCM Agent:** built a schedule-performance scoring module and a data-quality auditor that flagged 138 export issues (malformed PO numbers, duplicate item codes, auto-filled receipts); diagnosed and documented a root-cause data-pipeline defect (a weekly Store Input overwrite) causing misleading zero PR-to-PO conversion reads, and validated the agent's logic with a 28-case regression test suite.
- Built a staff lunch-booking and spend-tracking feature (Laravel/JavaScript) for Confidence Group employees entirely through AI-assisted development in Cursor Pro — directing implementation via prompt engineering in a stack outside prior hands-on experience.

RESEARCH PUBLICATIONS

Big Data Analysis for Forest Cover Prediction with PySpark

IEEE QPAIN 2026 - Chittagong, Bangladesh | DOI: [10.1109/QPAIN69676.2026.11546446](https://doi.org/10.1109/QPAIN69676.2026.11546446)

- Conducted large-scale forest cover prediction using PySpark distributed ML (Logistic Regression, Random Forest, SVM, MLP, CatBoost) with SMOTE class-imbalance handling; achieved 96.18% classification accuracy with CatBoost.
- Co-authors: Meherunnesa Hossain Ibnath, Sadia Sabrin Neha, Nusrat Jahan.

SNR-Aware Noise-Robust Keyword Spotting for Low-Resource Voice Command Recognition using Multi-Feature Hybrid CNN-BIGRU

IEEE SPICSCON 2026 - Conference | Status: Under Review (Revision Requested)

- Proposed an SNR-aware, noise-robust keyword-spotting model fusing multiple acoustic features for reliable low-resource voice command recognition under noisy conditions.
- Co-authors: Sadia Sabrin Neha, Moinul Hoque.

Uncertainty-Calibrated Learning Analytics for Learner Risk Support: Selective Prediction and Human-in-the-Loop Triage Using OULAD

IEEE MIET 2026 - Conference | Status: Under Review (Revision Requested)

- Developed an uncertainty-calibrated learning-analytics framework using selective prediction and human-in-the-loop triage to support early identification of at-risk learners on the OULAD dataset.
- Co-authors: Sadia Sabrin Neha, Moinul Hoque.

Time-Aware and Explainable Hybrid Ensemble Learning for PM10 Prediction Using Meteorological and Lag-Based Features

IEEE MIET 2026 - Conference | Status: Under Review (Revision Requested)

- Built a time-aware, explainable hybrid ensemble for PM10 air-quality prediction using meteorological and lag-based features.
- Co-authors: Sadia Sabrin Neha, Moinul Hoque.

Cross-Domain Toxicity Detection with Binary Label Harmonization: A Comparative Study of Fine-Tuned Transformers, Prompt-Based Language Models, and Hybrid Ensembles

IEEE SPICSCON 2026 - Conference | Status: Under Review (Revision Requested)

- Conducted a cross-domain toxicity-detection study with binary label harmonization, comparing fine-tuned transformers, prompt-based language models, and hybrid ensembles.
- Co-authors: Sadia Sabrin Neha, Moinul Hoque.

ACHIEVEMENTS

- 6th Place - RSBRU International Research Summit 2026: presented "Explainable Pneumonia Detection from Chest X-Ray Images Using Deep Learning and Grad-CAM" at Begum Rokeya University, Bangladesh.
- 8th Position (AUST) - AgentX: The AI Challenge 2025, NetCom Learning Bangladesh (Powered by Microsoft): competed in Bangladesh's largest AI prompting competition, covering Microsoft 365 Copilot, AI+ Prompt Engineering, and AI Agent Design/Creation; Prize-Giving Ceremony hosted at Ahsanullah University of Science and Technology.
- Campus Ambassador - ICT Olympiad Bangladesh (2022): represented AUST and promoted national-level ICT participation.

PROJECTS

FastAPI Resume Matcher | [GitHub](#)

FastAPI · SentenceTransformers · Scikit-learn · Python

- Developed a RESTful API using MiniLM sentence embeddings for semantic resume-job matching; added skill extraction, gap analysis, and Swagger UI testing.

ClinReasoner - Guideline-Grounded Clinical Decision Agent | [GitHub](#)

RAG · FAISS · Sentence-Transformers · Streamlit · Python

- Built a multi-domain clinical guideline RAG system using PyMuPDF and FAISS with rule-based risk detection and citation-backed recommendations.

ChurnBuster - Customer Churn Prediction & Retention Dashboard | [GitHub](#)

XGBoost · SHAP · Streamlit · Python

- Developed an end-to-end churn prediction dashboard with probability-based risk scores, cost-based threshold optimisation, and SHAP explainability.

Mobile Financial Service Campaign Performance Dashboard | [GitHub](#)

Power BI · Power Query · DAX · CSV Dataset · Data Visualization · KPI Reporting

- Designed an interactive Power BI dashboard analyzing revenue, cost, ROI, conversion rate, transactions, regional performance, and monthly trends using DAX KPIs and slicers.

Skin Disease Detection System | [GitHub](#)

CNN · FastAPI · Streamlit · Python · Kaggle Skin Disease Dataset

- Built a CNN-based multi-class skin disease classifier for 10 disease categories using 27k+ images; developed a FastAPI backend and Streamlit UI with image preprocessing and rule-based recommendations.

Uncertainty-Aware Retail Demand Forecaster | [Retail Repo](#) | [Uncertainty Repo](#)

Prophet · Streamlit · Scikit-learn · Python

- Built store-level demand forecasting with P50/P90 uncertainty estimates and inventory simulation; compared ordering strategies using MAE and cost-based metrics.

ECourse - Full-Stack E-Learning Platform | [GitHub](#)

Django · PostgreSQL · Bootstrap · Python

- Designed a Django e-learning platform with user auth, course creation, enrollment, progress tracking, role-based access, PostgreSQL schema, and responsive Bootstrap UI.

TECHNICAL SKILLS

Languages: Python, SQL, C, C++, JavaScript, PHP

Automation & AI Agents: Claude Skills / Agentic Workflows, LLM Orchestration, Cursor (AI pair-programming), Prompt Engineering, Workflow Automation, Process Automation, Data Pipeline Validation, Excel/Report Automation (openpyxl, python-docx, reportlab)

AI / ML: Machine Learning, Deep Learning, CNN, XGBoost, Scikit-learn, BanglaBERT, XLM-RoBERTa, mBERT, Prophet, SHAP, FAISS, LoRA/PEFT, PySpark, SMOTE

Research & NLP: LLM/NLP, Literature Review, Paper Analysis, RAG, Transformer Models, Experiment Documentation

Backend & Frameworks: FastAPI, Django (MTV/ORM), Streamlit, REST API Design

Data & Analytics: Pandas, NumPy, Matplotlib, Excel, Power BI, Power Query, DAX, KPI Reporting

Databases: PostgreSQL, MySQL

Tools & Platforms: Git & GitHub, Google Colab, VS Code, FastAPI/Uvicorn, ngrok, PyMuPDF, Render, Android Studio

CERTIFICATIONS & TRAINING

- AI+ Prompt Engineer Level 1™ - AI CERTs™
- Microsoft Azure AI Fundamentals (AI-900) - Microsoft
- Introduction to Generative AI & Image Generation - Google Cloud
- Machine Learning Specialization - Andrew Ng, Coursera
- Google Certified Educator: Level 1 - Google for Education
- Student Ambassador Technical Training Program - Microsoft
- Hands-on Workshop: FPGA Acceleration for ML Applications - AUST (Jan 2025)
- Hands-on Workshop: AWS, CNCF & GitHub - AUSTPIC (Aug 2025)

REFERENCES

Mohammad Mosharaf Hossain

DGM (Zonal Manager)

Grameen Bank

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